

The price impact of order book events

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PUBLICATION YEAR	2014 (arXiv version 3 reviewed: 2011)
JOURNAL / WORKING PAPER	Journal of Financial Econometrics, 12(1), 47–88; arXiv:1011.6402
DOI	10.1093/jjfinec/nbt003
RESEARCH AREA	Market microstructure; price impact; high-frequency equity data
SERIES	Market Microstructure Physics
PAPER TYPE	Mixed (empirical, with a stylised theoretical model and one limit theorem)
REVIEWED BY	Arthur Faugeron
DATE REVIEWED	9 October 2026

ANALYTICAL ABSTRACT

The paper asks which variable best explains short-horizon mid-price changes of equities when limit orders, market orders and cancellations are all allowed to matter. The authors define the order flow imbalance, a signed sum of changes in the best bid and ask queues that treats a market sell and a cancelled buy of equal size identically. Using one month of NYSE trades and quotes for 50 randomly chosen S&P 500 stocks, they regress 10-second mid-price changes on this imbalance in each half-hour window. The relation is linear, with an average R-squared of 65 percent, and its slope is inversely proportional to average market depth. Trade imbalance explains only 32 percent, and loses most of its significance once the order flow imbalance is included. A central limit argument then shows that, if order flow imbalance drives prices, an apparent square-root dependence of price changes on traded volume arises as an aggregation artefact. This review reconstructs the model and the limit theorem, audits the tables, and argues that the central relation is partly mechanical, contemporaneous and not identified causally, while remaining a robust descriptive law of the top of the book.

1 Introduction

1.1 Paper identification and classification

The paper is an empirical market microstructure study built around a stylised order book model and a short asymptotic result. It belongs to the field of market microstructure and, within it, to the subfield of price impact and liquidity measurement. The version read in full for this review is the arXiv text dated March 2011 (version 3, arXiv:1011.6402), which contains the abstract, introduction, model, estimation, trade-versus-quote comparison, conclusion, references and the data appendix, together with Tables 1 to 5 and Figures 1 to 6. The published version appeared in the *Journal of Financial Econometrics* in 2014; differences between the two texts cannot be established from the material read and are not assessed here. The series assignment, Market Microstructure Physics, reflects the physics-style programme of the paper: a single scaling variable, a linear response law, a power-law exponent close to unity, and a central-limit scaling argument.

1.2 Executive understanding

WHAT IS THE PAPER TRYING TO DISCOVER?

Whether the many order types that change a limit order book can be summarised, for the purpose of explaining short-run price moves, by one aggregate variable, and how the response of the price to that variable depends on market depth.

WHY DOES THE QUESTION MATTER?

Optimal execution models in the tradition of Almgren and Chriss (2000), Bertsimas and Lo (1998) and Obizhaeva and Wang (2013) take a price impact function as an input. The literature disagrees about its shape: linear, concave, square root, permanent, transient. A parsimonious and stable estimate of impact, together with an explanation of its intraday pattern, is directly usable by execution desks and by researchers who need a liquidity measure.

WHAT WAS KNOWN BEFORE, AND WHAT IS MISSING?

Earlier empirical work concentrated on trades and typically found impact to be concave in trade size (Hasbrouck 1991; Plerou et al. 2002; Torre and Ferrari 1997). Quotes had been studied by Eisler, Bouchaud and Kockelkoren (working paper 2009; Quantitative Finance 2012) and Hautsch and Huang (2009), but at the level of single events or by impulse responses. What was missing is an aggregate, easily computable measure that treats trades, limit orders and cancellations symmetrically, tested on a cross-section of stocks.

WHAT DOES THE PAPER PROPOSE, HOW DOES IT INVESTIGATE, AND WHAT DOES IT FIND?

It proposes the order flow imbalance, estimates a half-hourly linear regression for each of 50 stocks, relates the slope to depth, compares the variable with trade imbalance, and derives a square-root artefact for volume. It finds an average R-squared of 65 percent against 32 percent for trade imbalance, an estimated depth exponent close to one, and a price-volume exponent below one half. The findings matter because they would reduce price impact to a one-parameter, observable relation.

2 Research Problem and Hypothesis

2.1 Research question

PAPER The precise question is whether, over short intervals, the change in the mid-quote of a stock is explained by the net change in best-quote queue sizes, summarised in one variable, and whether the coefficient on that variable is inversely proportional to the average size of the best queues. A second question is whether trade imbalance and traded volume retain explanatory power once that variable is controlled for.

2.2 Hypotheses

PAPER The hypotheses are stated implicitly through the model and can be written as follows.

- H1: the mid-price change over a short interval is linear in the order flow imbalance, with no significant concavity.
- H2: the price impact coefficient satisfies $\beta \propto AD^{-\lambda}$ with $\lambda = 1$, where AD is average depth at the best quotes.
- H3: the effect of trades on prices is contained in the order flow imbalance, so the trade imbalance coefficient is zero in a joint regression.
- H4: the price-volume relation is an indirect one and is noisy, with an apparent exponent that can be below one half.

2.3 Existing literature and research gap

PAPER The authors cite three strands: impact of parent orders on proprietary data (Engle, Ferstenberg and Russell 2006; Almgren et al. 2005), impact of trades on public data (Hasbrouck 1991; Kempf and Korn 1999; Plerou et al. 2002), and limit order models (Bouchaud et al. 2004; Weber and Rosenow 2005; Stephens, Waelbroeck and Mendoza 2009). They argue that a trade-only focus ignores the roughly forty quote updates that occur for each trade in their data. **CRITIQUE** The gap is genuine, but the paper is closest to Eisler, Bouchaud and Kockelkoren, whose event-level model already gives market orders,

limit orders and cancellations linear impacts; the novelty lies mainly in the aggregation across events and time and in the cross-sectional breadth, not in the idea that quote events matter.

2.4 Claimed contribution

The claimed contribution is a single-parameter linear model, a depth explanation of intraday seasonality, and a demonstration that volume is a derived variable. The first and the third are supported descriptively; the second is supported but close to an accounting relation (Section 7); none is a causal identification of impact.

3 Theoretical Framework and Mechanism

3.1 The order flow imbalance

PAPER Observe the best bid price P_n^B , bid queue size q_n^B , best ask price P_n^A and ask queue size q_n^A at each quote update n . The contribution of the n -th event to the imbalance of supply and demand is

$$e_n = \mathbb{I}_{\{P_n^B \geq P_{n-1}^B\}} q_n^B - \mathbb{I}_{\{P_n^B \leq P_{n-1}^B\}} q_{n-1}^B - \mathbb{I}_{\{P_n^A \leq P_{n-1}^A\}} q_n^A + \mathbb{I}_{\{P_n^A \geq P_{n-1}^A\}} q_{n-1}^A. \quad (1)$$

e_n	Signed contribution of event n ; positive when demand increases or supply decreases.
q_n^B, q_n^A	Sizes of the best bid and best ask queues after event n , in the units reported by the data.
$\mathbb{I}_{\{\cdot\}}$	Indicator function; the price comparisons select which queue size enters.

Equation (1) has four cases for the bid side. If the bid price is unchanged, then $\mathbb{I}_{\{P_n^B \geq P_{n-1}^B\}} = \mathbb{I}_{\{P_n^B \leq P_{n-1}^B\}} = 1$ and the bid term collapses to $q_n^B - q_{n-1}^B$, the net shares added (a limit buy) or removed (a market sell or a cancellation). If the bid price rises, only the first indicator is active and the term equals q_n^B , the whole new queue at the better price. If the bid price falls, only the second indicator is active and the term equals $-q_{n-1}^B$, the whole old queue that disappeared. The ask side mirrors this with opposite signs. The key economic feature is that a market sell of size x and a cancellation of a buy order of size x both reduce q^B by x and so enter identically, which is the statement that the book reacts to the net change in resting liquidity and not to the label of the order. Over an interval $[t_{k-1}, t_k]$ the imbalance and the tick-denominated mid-price change are

$$\text{OFI}_k = \sum_{n=N(t_{k-1})+1}^{N(t_k)} e_n, \quad \Delta P_k = \frac{P_k - P_{k-1}}{\delta}, \quad (2)$$

where $N(t)$ counts the events up to t , P_k is the mid-quote at t_k and δ is the tick size (one cent in the data). **CRITIQUE** A property that the paper does not discuss is that e_n is built from the same quote updates that move the mid-quote: whenever the best bid rises, $e_n = q_n^B > 0$ by construction. The regressor therefore contains the dependent variable's own jumps, a point taken up in Section 7.

3.2 Stylised order book and the linear law

PAPER Assume (i) every price level has D shares, including levels beyond the best quotes, and (ii) limit orders and cancellations arrive only at the best bid and ask. Over a short interval let L^b, C^b, M^s denote limit buy additions, buy cancellations and market sells at the bid, and L^s, C^s, M^b the analogous ask-side quantities. The net change in the bid queue is $L^b - C^b - M^s$. Because each full queue of D shares is one tick, the bid and ask prices move by

$$\Delta P^b = \left\lceil \frac{L^b - C^b - M^s}{D} \right\rceil, \quad \Delta P^a = - \left\lceil \frac{L^s - C^s - M^b}{D} \right\rceil. \quad (3)$$

Averaging bid and ask moves gives the mid-price change, and the derivation of the linear law is the following chain, which the paper compresses:

$$\begin{aligned} \Delta P &= \frac{1}{2} \Delta P^b + \frac{1}{2} \Delta P^a = \frac{1}{2} \left\lceil \frac{L^b - C^b - M^s}{D} \right\rceil - \frac{1}{2} \left\lceil \frac{L^s - C^s - M^b}{D} \right\rceil \\ &= \frac{(L^b - C^b - M^s) - (L^s - C^s - M^b)}{2D} + \varepsilon = \frac{\text{OFI}}{2D} + \varepsilon, \end{aligned} \quad (4)$$

where $\text{OFI} = L^b - C^b - M^s - L^s + C^s + M^b$ is the grouping of the e_n by order type and ε collects the two truncation errors, each bounded by one half in absolute value. The equation says that the mid-price moves by one tick for every $2D$ shares of net imbalance. It predicts a parameter-free slope $\beta = 1/(2D)$, so impact is inversely proportional to depth, and it predicts that every order type has the same impact per share. **CRITIQUE** The ceiling function is not innocuous: under assumption (ii) a net addition $0 < L^b - C^b - M^s < D$ at the existing best bid cannot move the bid price, whereas the ceiling returns $+1$. The formula is therefore exact only for net removals, and for net additions it requires price-improving orders that assumption (ii) excludes. The stylised model is thus a heuristic that motivates the sign and the $1/D$ scaling, not a derivation of linearity.

3.3 Statistical specification

PAPER Because real books have uneven depth profiles, hidden orders and deeper-level activity, the authors assume that equation (4) holds only statistically,

$$\Delta P_k = \beta \text{OFI}_k + \varepsilon_k, \quad \beta_i = c \text{AD}_i^{-\lambda} + \nu_i, \quad (5)$$

with depth measured over a half-hour window $[T_{i-1}, T_i]$ by

$$\text{AD}_i = \frac{1}{2(N(T_i) - N(T_{i-1}) - 1)} \sum_{n=N(T_{i-1})+1}^{N(T_i)} (q_n^B + q_n^A). \quad (6)$$

In the third equation of the system the paper writes the depth law as $\beta_i = c \text{AD}_i^\lambda + \nu_i$ in the text extracted from the electronic version, but the surrounding argument, equation (12) and the reported estimates of $\hat{\lambda}$ near $+1$ all imply the inverse relation, so the exponent is read here as $-\lambda$. The stylised model corresponds to $\lambda = 1$ and $c = 1/2$. Taking variances in the first equation of (5) gives the link used for

intraday volatility,

$$\text{var}[\Delta P_k]_i = \beta_i^2 \text{var}[\text{OFI}_k]_i + \text{var}[\varepsilon_k]_i, \quad (7)$$

which assumes $\text{cov}(\text{OFI}_k, \varepsilon_k) = 0$ within the window. The meaning is that the variance of 10-second price changes is the product of squared impact and the variance of the imbalance: price volatility is high when either order flow is volatile or the book is thin.

3.4 Proposition 1: the square-root artefact

PAPER Let $\text{VOL}(T) = \sum_{n \leq N(T)} w_n$ be traded volume in $[0, T)$, with w_n the size of any trade at quote n and zero otherwise. Assume that (1) $N(T)/T \rightarrow \Lambda$; (2) the ε_i are i.i.d. with finite variance σ^2 ; (3) the w_i are i.i.d. with finite mean $\mu\pi$, where π is the proportion of events that are trades and μ the mean trade size. The proposition states a normal limit for the ratio of the imbalance to the square root of volume. The electronic text from which this review was transcribed drops radical signs, so the following derivation was redone from the stated assumptions and gives the form that is dimensionally consistent:

$$\frac{\sqrt{\mu\pi}}{\sigma} \frac{\text{OFI}(T)}{\sqrt{\text{VOL}(T)}} \Rightarrow \xi \sim \mathcal{N}(0, 1) \quad (T \rightarrow \infty). \quad (8)$$

The proof has two steps. First, the law of large numbers applied to the trade sizes gives $\text{VOL}(T)/N(T) = \sum_{i \leq N(T)} w_i/N(T) \rightarrow \mu\pi$ almost surely. Second, the central limit theorem with a random number of terms (Anscombe's lemma, since $N(T)$ is random but tends to infinity) gives $\text{OFI}(T)/(\sigma\sqrt{N(T)}) \Rightarrow \xi$. Combining the two by Slutsky's theorem,

$$\frac{\sqrt{\mu\pi}}{\sigma} \frac{\text{OFI}(T)}{\sqrt{\text{VOL}(T)}} = \underbrace{\frac{\text{OFI}(T)}{\sigma\sqrt{N(T)}}}_{\Rightarrow \xi} \cdot \underbrace{\sqrt{\frac{\mu\pi N(T)}{\text{VOL}(T)}}}_{\rightarrow 1} \Rightarrow \xi. \quad (9)$$

Rearranged, $\text{OFI}(T) \approx \xi(\sigma/\sqrt{\mu\pi})\sqrt{\text{VOL}(T)}$. Substituted into the first equation of (5) this gives $\Delta P_k = \theta_k \sqrt{\text{VOL}_k} + \varepsilon_k$ with $\theta_k = \beta_i \xi_k \sigma / \sqrt{\mu\pi} \sim \mathcal{N}(0, \beta_i^2 \sigma^2 / (\mu\pi))$, and the generalisation $\Delta P_k = \theta_k \text{VOL}_k^H + \varepsilon_k$ for exponents $0 < H < 1$ when the independence or finite-variance assumptions fail. The economic reading is that volume is a proxy for the number of events, and the number of events scales the size of the net imbalance as its square root because buy and sell contributions largely cancel. The coefficient θ_k is random with mean zero, so a regression on $\sqrt{\text{VOL}}$ cannot recover a stable positive slope; this is why the authors do not recommend the model. The proposition omits the assumption $\mathbb{E}[\varepsilon_i] = 0$, without which the stated normal limit does not hold, and it ignores the sign of the imbalance, so the artefact concerns $|\Delta P_k|$ and not ΔP_k .

INFERENCE A first-order calculation shows what the proposition implies for the log-log exponent actually estimated. Let $\log |\Delta P_k| = \log \beta + \frac{1}{2} \log N_k + \log \sigma + \log |\xi_k|$ and $\log \text{VOL}_k = \log N_k + \log(\mu\pi) + \eta_k$, where η_k is the relative fluctuation of volume around $\mu\pi N_k$ with variance approximately $(1 + \text{CV}_w^2)/(\pi N_k)$ and CV_w is the coefficient of variation of trade size. If $\log |\xi_k|$ and η_k are independent of

$\log N_k$, the least-squares slope of the first on the second tends to

$$H \approx \frac{1}{2} \frac{\text{var}(\log N)}{\text{var}(\log N) + (1 + \text{CV}_w^2)/(\pi \bar{N})}. \quad (10)$$

This is attenuation bias from measuring the event count by volume: the exponent equals one half only when the number of events varies across intervals far more than the sampling noise in volume, and it falls to zero when the window contains a nearly constant number of events. The estimated $\hat{H}$ of 0.18 on average (Table 5) is therefore compatible with the mechanism and does not by itself confirm it. Section 8 illustrates this with a simulation.

4 Data and Empirical Strategy

4.1 Data

PAPER The data are one calendar month, April 2010, of NYSE Trades and Quotes (TAQ) consolidated quotes and trades for 50 stocks drawn by a random number generator from the S&P 500 constituents of that month (constituents from Compustat; data through Wharton Research Data Services). There are 21 trading days. Each quote record has a timestamp rounded to the second, bid and ask prices and sizes, and an exchange flag; each trade has a timestamp, price and size. The paper reports 223,232 best-quote updates and 4,552 trades per stock-day on average (Table 1), a ratio of roughly 40 to 1 reported in the text, which motivates the focus on quotes. Table 1 reproduces selected rows of Table 1 of the paper to show the range of the sample.

Table 1. Selected descriptive statistics of the 50-stock sample (rows of Table 1 of the paper)

Stock	Price	Volume	Quotes	Trades	Spread	Depth
BAC	18.43	164,550,168	1,529,395	15,008	1	3,208
SLB	67.94	9,476,060	440,839	10,286	1	39
AXP	45.21	8,678,723	559,701	7,748	1	79
AZO	179.03	243,197	43,682	1,081	9	7
CME	322.83	418,955	38,504	1,412	31	5
WIN	11.03	2,508,830	104,887	2,937	1	798
Grand mean, 50 stocks	51.75	7,512,376	223,232	4,552	2	227

Source: Cont, Kukanov and Stoikov (arXiv version 3), Table 1; selected rows copied from the paper (official source). Columns: average mid-price in dollars; daily volume in shares; daily best-quote updates; daily trades; average spread in cents; average size at the best quotes in the units of the paper. Counts are daily and after filtering.

4.2 Construction of variables

Quotes are kept between 9:30 and 16:00 with positive prices and sizes and a regular quote mode; trades are kept with positive price and size, correction indicator at most two, and a set of excluded sale conditions. The national best bid and offer (NBBO) is rebuilt at every quote update by keeping the latest best quote of each exchange, taking the highest bid and lowest ask, and summing the sizes at the NBBO prices across exchanges. Trades are matched to the NBBO with a quote test: a trade at or above the ask is a buy, at or below the bid is a sell, trades inside the spread are left unmatched, and a trade is matched to the earliest quote within one second. The authors also state, without reporting the numbers, that the tick test gives virtually the same results. Quotes in the highest 5 percent of each stock's own spread distribution are removed. The time grid is 10 seconds, so a half-hour window holds about 180 observations; 21 days with 13 windows give 273 windows per stock.

4.3 Estimation strategy

PAPER For each stock and half-hour window the authors estimate by ordinary least squares

$$\Delta P_k = \hat{\alpha}_i + \hat{\beta}_i \text{OFI}_k + \hat{\varepsilon}_k, \quad (11)$$

with White heteroskedasticity-consistent standard errors, and a variant with an extra term $\hat{\gamma}_{Q,i} \text{OFI}_k |\text{OFI}_k|$ to detect concavity. Output is averaged across windows. The depth law is then estimated across windows for each stock by a log-log regression and a nonlinear-in-parameters linear regression,

$$\log \hat{\beta}_i = \hat{\alpha}_{L,i} - \hat{\lambda} \log \text{AD}_i + \hat{\varepsilon}_{L,i}, \quad \hat{\beta}_i = \hat{\alpha}_{M,i} + \hat{c} \text{AD}_i^{\hat{\lambda}} + \hat{\varepsilon}_{M,i}, \quad (12)$$

with Newey-West standard errors because residuals are autocorrelated. The comparison with trades uses the trade imbalance $\text{TI}_k = \sum b_n - \sum s_n$ (buyer-initiated minus seller-initiated volume) in three regressions,

$$\Delta P_k = \hat{\alpha}_i + \hat{\beta}_i \text{OFI}_k + \hat{\varepsilon}_k, \quad (13)$$

$$\Delta P_k = \hat{\alpha}_{T,i} + \hat{\beta}_{T,i} \text{TI}_k + \hat{\varepsilon}_{T,k}, \quad (14)$$

$$\Delta P_k = \hat{\alpha}_{D,i} + \hat{\theta}_{O,i} \text{OFI}_k + \hat{\theta}_{T,i} \text{TI}_k + \hat{\varepsilon}_{D,k}, \quad (15)$$

and the volume comparison in absolute values, with exponents pre-estimated in each window from $\log |\Delta P_k| = \log \bar{\theta}_i + \hat{H}_i \log \text{VOL}_k + \log \hat{\xi}_k$,

$$|\Delta P_k| = \hat{\alpha}_{O,i} + \hat{\beta}_{O,i} |\text{OFI}_k| + \hat{\varepsilon}_{O,k}, \quad (16)$$

$$|\Delta P_k| = \hat{\alpha}_{V,i} + \hat{\beta}_{V,i} \text{VOL}_k^{\hat{H}_i} + \hat{\varepsilon}_{V,k}, \quad (17)$$

$$|\Delta P_k| = \hat{\alpha}_{W,i} + \hat{\phi}_{O,i} |\text{OFI}_k| + \hat{\phi}_{V,i} \text{VOL}_k^{\hat{H}_i} + \hat{\varepsilon}_{W,k}. \quad (18)$$

4.4 Where does the identifying variation come from?

CRITIQUE The variation is the within-window, 10-second co-movement of two quantities built from the same quote stream. No exogenous shifter of order flow is used, so the design identifies a conditional association, not a causal impact. Cross-sectional variation in depth, which identifies λ , comes from the permanent differences between stocks and from intraday seasonality, both of which move together with tick size, price level and spread.

4.5 Data limitations and biases

- Selection and sample bias: 50 large-cap S&P 500 stocks in one month. There is no small-cap or low-liquidity stock, no stress period, and no change of regime. Survivorship is not a concern within a month, but external validity is.
- Measurement error: timestamps are rounded to the second, so the order of events within a second is unknown, and the consolidated NBBO sums sizes across exchanges, which can mix stale and live quotes. Hidden and iceberg liquidity is not observed, and the authors acknowledge it as a source of error. The units of quote size are not stated explicitly in the paper; TAQ quote sizes are conventionally reported in round lots of 100 shares, which would change the interpretation of every slope by a constant factor but not any statistic reported.
- Look-ahead bias: absent from the regressions because all variables are contemporaneous, but for the same reason the model has no forecasting content.
- Data mining: one month, one specification family, 50 stocks drawn at random; the main risk is not data mining but the small number of independent time periods.
- Trade classification: the quote test leaves trades inside the spread unclassified, which contributes error to TI_k and not to OFI_k , and therefore tilts the comparison in favour of the quote-based variable.

5 Results

5.1 Linear price impact of order flow imbalance

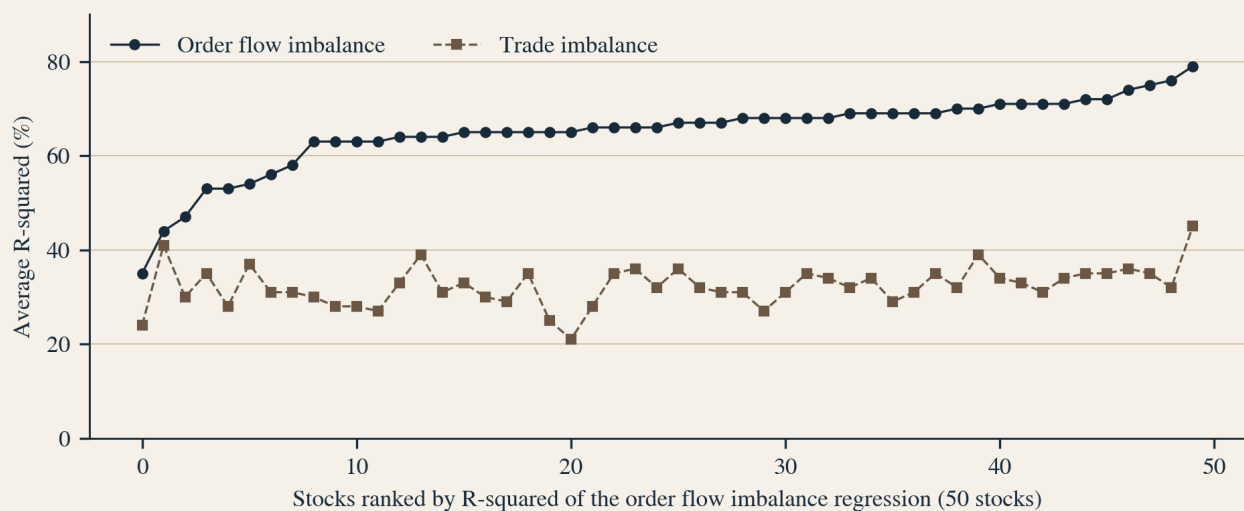
EVIDENCE Table 2 reproduces selected rows of Table 2 of the paper. Averaged across stocks and windows, $\hat{\beta}$ has a t -statistic of 11.5 and is significant at the 5 percent level in 97 percent of windows; the intercept is insignificant in 94 percent of windows (it is significant in 6 percent, close to the nominal rate); the average R-squared is 65 percent. The quadratic term $\hat{\gamma}_Q$ is significant in only 9 percent of windows and raises the R-squared from 65 to 68 percent. The linearity hypothesis H1 is therefore supported, in the sense that no systematic concavity is found at the 10-second scale. Figure 1 shows the dispersion of fit across the sample.

Table 2. Price changes on order flow imbalance, selected rows of Table 2 of the paper

Stock	$\hat{\beta}$ (ticks per unit)	$t(\hat{\beta})$	R^2	$\{\beta \neq 0\}$	$t(\hat{\gamma}_Q)$	$\{\gamma_Q \neq 0\}$
BAC	0.0002	18.36	79%	100%	0.01	8%
SLB	0.0198	16.76	76%	100%	-1.15	1%
AXP	0.0082	13.87	69%	100%	-0.88	5%
AZO	0.1619	6.39	47%	97%	-0.89	3%
CME	0.6262	4.98	35%	94%	-0.99	2%
WIN	0.0009	3.12	44%	60%	0.76	15%
Average, 50 stocks	0.0398	11.47	65%	97%	-0.28	9%

Source: Cont, Kukanov and Stoikov (arXiv version 3), Table 2, selected rows copied from the paper (official source). Dependent variable: 10-second change in the mid-quote in ticks. Columns marked with braces give the share of the 273 half-hour windows in which the coefficient is significant at 5 percent.

EVIDENCE Interpretation: the table answers whether a linear one-regressor model fits within windows. The key coefficient is $\hat{\beta}$, and the table fails to show whether it is causal. In economic terms, for Schlumberger an imbalance of 100 quote units moves the mid-quote by about $0.0198 \times 100 \approx 2$ ticks, that is two cents on a price of 67.94 dollars, or about 0.3 basis point; for Bank of America the same imbalance moves it by $0.0002 \times 100 = 0.02$ tick. For a thin, high-priced stock such as CME the impact per unit is 3,000 times larger than for Bank of America, so the cross-sectional dispersion of $\hat{\beta}$ is dominated by depth. **CRITIQUE** The units of OFI determine whether these figures refer to shares or to round lots, so they are stated per unit and not per share.

Figure 1. Average R-squared of mid-price changes regressed on order flow imbalance and on trade imbalance, by stock

Source: values copied from Table 4, Panel A of the paper (official source), plotted by the reviewer. 50 S&P 500 stocks, April 2010, 10-second intervals; each point is the average over 273 half-hour windows. Stocks are ranked by the R-squared of the order flow imbalance regression.

EVIDENCE Figure 1 shows the visible relation. The order flow imbalance line lies above the trade imbalance line for every stock, with the narrowest gap at Windstream (44 percent against 41 percent, where both fits are poor), and the two series are not parallel: trade imbalance R-squared is nearly flat between 21 and 45 percent while the order flow imbalance R-squared ranges from 35 to 79 percent. The figure is descriptive and carries no causal content; the unusual observations are CME, Autozone and Windstream, which have wide spreads or very thin or very thick queues.

5.2 Depth and the price impact coefficient

EVIDENCE Table 3 reproduces selected rows of Table 3 of the paper. The fitted exponent has a grand mean of $\hat{\lambda} = 0.98$ and the hypothesis $\lambda = 1$ cannot be rejected for 35 of 50 stocks; the average constant is $\hat{c} = 0.45$ against the stylised value 0.5. The R-squared of the log-log regression averages 74 percent. The three poor fits (Apollo, Autozone, CME) are stocks with wide spreads and very thin queues, where the paper concedes that deeper levels and hidden orders probably dominate. The table supports H2 for liquid stocks and does not establish it for thin ones.

Table 3. Price impact coefficient and market depth, selected rows of Table 3 of the paper

Stock	$\hat{c}$	$\hat{\lambda}$	95% interval for λ	R^2 of (12)	Squared correlation, fitted
AMD	0.23	0.94	0.86 to 1.02	78%	86%
AXP	0.14	0.83	0.77 to 0.89	84%	76%
BAC	0.27	0.96	0.90 to 1.03	76%	87%
BK	0.21	0.92	0.88 to 0.95	93%	91%
APOL	0.27	0.36	−0.32 to 1.04	2%	30%
AZO	0.39	0.67	0.41 to 0.92	13%	17%
CME	1.27	0.50	0.01 to 0.99	2%	4%
Grand mean, 50 stocks	0.45	0.98	0.88 to 1.08	74%	75%

Source: Cont, Kukanov and Stoikov (arXiv version 3), Table 3, selected rows copied from the paper (official source). Estimates from 273 half-hour values of the impact coefficient per stock; Newey-West standard errors. The last column is the squared correlation between the estimated coefficient and its fitted value.

INFERENCE The reviewer's own calculation from the published Tables 1 and 2 provides an independent check of the depth law across stocks, with the caveat that it combines averages of different objects (the average of half-hourly coefficients from Table 2 and average best-quote depth from Table 1). Regressing $\log \hat{\beta}$ on $\log(\text{depth})$ over the 50 stocks gives a slope of -1.06 with an R-squared of 0.98, and the product of the coefficient and depth has a mean of 0.85 and a median of 0.79 across stocks, against the stylised value $\beta D = 0.5$. Figure 2 displays these points.

Figure 2. Average impact coefficient against average best-quote depth, 50 stocks (logarithmic axes)

Source: reviewer's calculation from Tables 1 and 2 of the paper (inputs official, regression generated by the reviewer). April 2010, 10-second intervals. The dashed line is the stylised-model prediction $1/(2D)$; the continuous line is the fitted log-log regression. Units of depth and imbalance as in the paper.

EVIDENCE Figure 2 shows an almost exact inverse proportionality across three orders of magnitude of depth, with most stocks above the stylised line by a factor of about one and a half, and CME above it by a factor of six. The relation is not causal evidence for the model: since mid-price changes are measured in whole ticks and queue sizes scale with the stock's liquidity, a unit-scaling relation of this kind is expected from any mechanism by which depletion of a queue moves the quote, and the fit is therefore a weak test of the stylised model's specific content. The coefficient is at least higher than $1/(2D)$, which is consistent with the paper's remark that deeper levels and hidden liquidity make the effective D at the best quote an imperfect proxy.

5.3 Intraday patterns

PAPER Averaged by half-hour of the day and across stocks, depth at the open is about half its daily mean and the impact coefficient about twice its daily mean; impact at the open is about five times its level at the close. Variance of price changes is high at the open because impact is high, even though the variance of the imbalance peaks at the close; equation (7) with the estimated $\hat{\beta}_i^2 \text{var}[\text{OFI}_k]_i$ tracks the variance of ΔP_k in Figure 6 of the paper. **CRITIQUE** The figures (Figures 5 and 6 of the paper) are averages over only 21 days and are not accompanied by dispersion bands, so the statistical precision of the open-to-close ratio is not established. The statement that price impact and information asymmetry are "two sides of the same coin" is an interpretation, not a test, since no information-asymmetry measure is estimated.

5.4 Order flow imbalance against trade imbalance

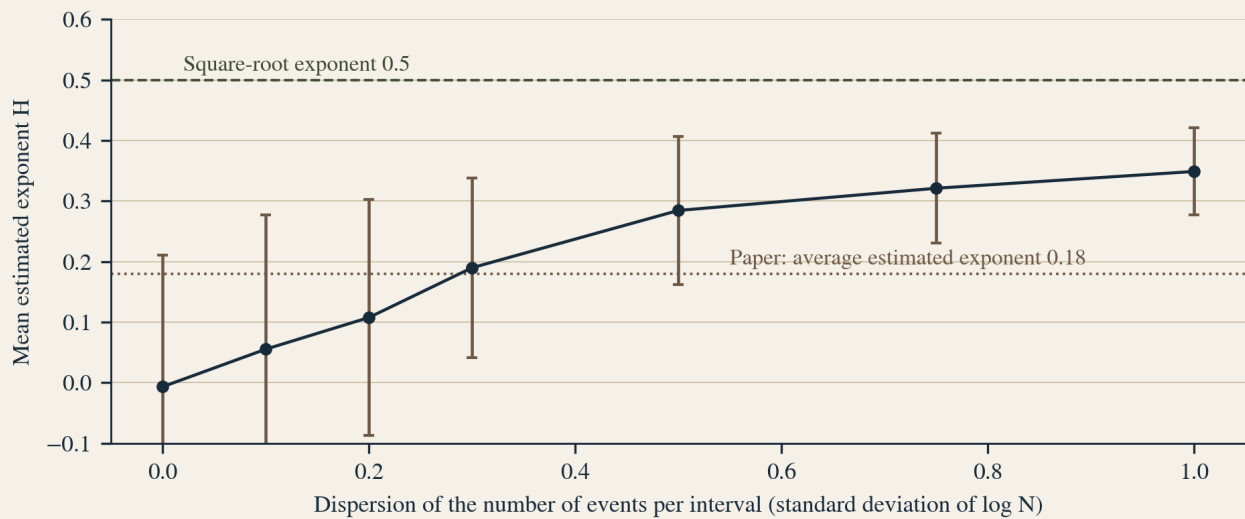
EVIDENCE Table 4 reproduces the grand means and Panel B of Table 4. Panel A of the paper shows an average R-squared of 65 percent for the imbalance of orders and 32 percent for the imbalance of trades; adding trade imbalance to the order flow imbalance raises R-squared from 65 to 67 percent, and the t -statistic of $\hat{\theta}_T$ falls from 4.18 (alone) to 1.16, with θ_T significant in 31 percent of windows against about 5 percent expected by chance. The R-squared ratio is about 2.0 (the reviewer's average of stock-level ratios is 2.04). Hypothesis H3, that the trade coefficient is zero once the imbalance is included, is therefore supported as a statement about economic magnitude and rejected as a statement about statistical zero.

Table 4. Order flow imbalance and trade imbalance, grand means of Table 4 of the paper

Specification	R^2 OFI alone	R^2 TI alone	R^2 both	$t(\hat{\theta}_O)$	$t(\hat{\theta}_T)$	$\{\theta_T \neq 0\}$
Panel A. Mid-price changes, 10 seconds	65%	32%	67%	8.49	1.16	31%
Panel B. Trade-price changes, 2 trades	14%	1%	15%	14.17	-2.58	54%
Panel B. Trade-price changes, 5 trades	38%	8%	39%	16.85	-0.20	9%
Panel B. Trade-price changes, 10 trades	51%	13%	51%	14.97	0.57	9%

Source: Cont, Kukanov and Stoikov (arXiv version 3), Table 4; Panel A is the grand mean over 50 stocks and Panel B the average over five stocks picked at random, copied from the paper (official source). Panel B uses changes in transaction prices over L trades.

EVIDENCE Panel B is informative in two ways. The gap between the two R-squared values is much wider for trade-price changes over short windows (14 against 1 percent over two trades), which is consistent with the order flow imbalance capturing price-relevant quote revisions that trades alone miss. In addition, the trade imbalance coefficient is significantly negative at the two-trade horizon (average $t = -2.58$, significant in 54 percent of windows), which the paper does not discuss. **INFERENCE** A negative sign on trade imbalance conditional on the order flow imbalance is the signature of the bid-ask bounce, because trade prices alternate between bid and ask while the imbalance already contains the quote shift.

Figure 3. Estimated price-volume exponent as a function of the dispersion of the number of events per interval (simulation)

Source: generated by the reviewer; not from the paper. Simulated intervals of 180 observations; order book events i.i.d. normal with standard deviation 200; trade probability 0.05; lognormal trade sizes with mean 300 and log standard deviation 1; mean event count 400; impact coefficient 0.02; 300 replications; points are mean estimates and bars one standard deviation across replications. These parameter values are the reviewer's assumptions.

5.5 Does volume move prices?

EVIDENCE Table 5 reproduces the grand mean of Table 5. The exponent $\hat{H}$ averages 0.18, with a cross-window standard deviation also of 0.18 and wide stock-level dispersion (from -0.17 for Boston Scientific, which has an unusually unstable estimate, to 0.49 for CME). In the absolute-value regressions, the imbalance explains 58 percent of the magnitude of price moves and volume only 23 percent; in the joint regression, the volume coefficient has a t -statistic of 2.3 and is significant in 67 percent of windows, the imbalance coefficient has $t = 8.3$, and R-squared rises only from 58 to 61 percent. The statement in the paper that only $|\text{OFI}|$ remains significant in the multiple regression is therefore too strong; the accurate statement is that the volume coefficient shrinks by roughly 60 percent in t -statistic terms (from 6.0 to 2.3) and adds three points of R-squared.

Table 5. Order flow imbalance and traded volume as explanations of absolute price changes, grand mean of Table 5

Row	$\hat{H}$	R^2 OFI	R^2 volume	R^2 both	$t(\hat{\phi}_O) / t(\hat{\phi}_V)$	$\{\phi_V \neq 0\}$
Grand mean, 50 stocks	0.18	58%	23%	61%	8.3 / 2.3	67%

Source: Cont, Kukanov and Stoikov (arXiv version 3), Table 5, grand mean copied from the paper (official source). t -statistics in the last two columns are for the joint regression; the stand-alone t -statistics are 9.8 for the imbalance and 6.0 for volume.

6 Robustness and Additional Tests

CRITIQUE The robustness evidence in the paper is concentrated in short passages and is mostly not tabulated. Table 6 states each concern, the test reported, the result, and the reviewer's assessment of whether the test resolves the concern.

Table 6. Robustness tests: concern, test, result and assessment

Concern	Test reported	Result	Does it resolve the concern?
Heteroskedastic residuals	White standard errors in each window	$\hat{\beta}$ significant in 97% of windows	Partly: corrects standard errors but not the mechanical link between regressor and dependent variable
Non-linearity	Quadratic term $\text{OFI}_k \text{OFI}_k $	R^2 from 65% to 68%; $\hat{\gamma}_Q$ insignificant in most windows (significant in 9%)	Yes for 10-second mid-price changes; concavity does appear for trade prices
Choice of time scale	Intervals from 10 quote updates to 10 minutes on a subsample	Fit rises with the interval, other results unchanged	Partly: the results are not reported, so magnitudes cannot be audited
Choice of exchange	Single exchange by flag instead of NBBO	Similar results	Partly: not reported
Trade classification	Tick test instead of quote test on a subsample	Virtually the same	Partly: not reported, and trades inside the spread are unclassified
Mid-price versus trade price	Repeat (8a) to (8c) with transaction price changes over L trades	Order flow imbalance still dominates; R^2 much lower at small L	Partly: five stocks only, and trade-time sampling induces its own biases
Seasonality	Half-hour windows with separate coefficients	Intraday patterns in $\hat{\beta}$ follow depth	Yes for the level of impact; the seasonality in the variance of the imbalance is not modelled
Autocorrelation of errors in the depth law	Newey-West standard errors	$\lambda = 1$ not rejected for 35 of 50 stocks	Partly: lags and bandwidth not reported
Outliers and spreads	Removal of the widest 5% of spreads per stock	Not reported separately	Unclear: affects the very stocks (CME, Autozone) that fit poorly

Source: reviewer's assessment of the robustness statements in Sections 3 and 4 and in the Appendix of the paper.

CRITIQUE None of the tests addresses the main threats: reverse causality (quotes react to price changes), omitted common drivers (news), the mechanical link between OFI and the quote changes that define ΔP , and out-of-sample stability. There is no placebo test, for instance a regression of price changes on the imbalance of a different stock or of a lagged window, and no subsample by time or volatility regime.

7 Critical Evaluation

7.1 What the paper establishes

ESTABLISHED

Over 10-second intervals, in this sample of 50 large-cap stocks in April 2010, the mid-price change and the order flow imbalance are strongly and positively associated, with an average R-squared of 65 percent and a coefficient significant in 97 percent of windows. The order flow imbalance explains about twice as much of the variation as trade imbalance does. The impact coefficient is approximately inversely proportional to average depth across stocks (the exponent $\hat{\lambda}$ averages 0.98).

SUPPORTED BUT CONDITIONAL

Linearity of impact at the 10-second scale (the quadratic term adds three points of R-squared) depends on the aggregation, the exclusion of the widest spreads, and the use of mid-prices. The intraday pattern in impact follows the pattern in depth, but it rests on 21 days of data. The claim that trades add nothing once the imbalance is included holds in magnitude and not in statistical significance.

SUGGESTIVE

That the apparent square-root dependence on volume is an aggregation artefact. The proposition is a valid asymptotic statement under its assumptions, and the estimated exponent of 0.18 is consistent with it (equation 10), but the paper does not test the assumptions of independent identically distributed event contributions with finite variance, which are contradicted by known long memory in order flow.

NOT ESTABLISHED

That order flow imbalance causes price changes; that it forecasts them; that the findings hold outside large-cap US equities in 2010 or in stressed markets; that price impact and information asymmetry are the same phenomenon; that the stylised model, as distinct from the unit scaling of ticks and queues, explains the depth law.

7.2 Strengths

STRENGTHS RECOGNISED BY THE AUTHORS

- Parsimony: one regressor and one parameter per window, estimable from freely accessible TAQ data. This matters because it makes the measure usable by practitioners and replicable.
- Robustness across stocks and scales: the R-squared exceeds 44 percent for every stock, and the relation is said to be stable across time scales.

- A unified treatment of market orders, limit orders and cancellations, which respects the fact that quote updates outnumber trades by about forty to one.

STRENGTHS IDENTIFIED INDEPENDENTLY

- Transparent tables: every stock is reported, and the share of windows in which each coefficient is significant is given, so heterogeneity is visible and not hidden by pooling. This allows the reader, as in the present review, to detect that the volume coefficient is not insignificant.
- A valid limit theorem with a clear message: the proposition gives a quantitative reason why volume-based impact models inherit concavity, and equation (10) shows that the proposition predicts the low exponents actually estimated.
- Honest treatment of failure cases: the paper names the three stocks with poor depth fits and offers a mechanism.
- The half-hour window approach adapts to intraday non-stationarity in a way that a single daily regression cannot.

7.3 Weaknesses and critique

SINGLE MOST IMPORTANT WEAKNESS

CRITIQUE Problem. The regressor and the dependent variable are built from the same quote updates, and the design has no source of exogenous variation. When the best bid rises, equation (1) assigns $e_n = q_n^B > 0$; the price jump itself therefore enters OFI. **Why it matters.** Part of the 65 percent R-squared is definitional, and the comparison with trade imbalance, which is not built from the quote changes that define ΔP , is tilted in favour of OFI. The word impact then overstates what is measured, namely a contemporaneous accounting relation between queue changes and the quote. **Potential bias.** Upward bias in R-squared and in $\hat{\beta}$, of unknown size. **How it could be tested.** Recompute OFI using only events that leave both best prices unchanged (queue changes at constant prices) and re-estimate; and regress ΔP_k on lagged OFI_{k-1} , which removes the mechanical link.

OTHER WEAKNESSES

- Reverse causality and common drivers. **Problem:** news moves prices and order flow together, and price changes induce quote revisions (replenishment, repricing). **Why it matters:** impact cannot be separated from information. **Potential bias:** $\hat{\beta}$ includes the adverse-selection component and cannot be used as a cost of trading a parent order. **Test:** an instrument or natural experiment that shifts queue sizes independently of information, such as a tick-size change or a technical outage at one venue.
- Near-identity of the depth law. **Problem:** ticks are discrete while queue sizes scale with a stock's liquidity, so $\beta \propto 1/D$ follows from unit scaling. **Why it matters:** the high R-squared of the log-log fit (0.98 in the reviewer's cross-section) is a weak test. **Potential bias:** overstated confidence in the model. **Test:** compare with a placebo in which depth is replaced by a rescaled variable unrelated to the book, for instance average price or average daily volume.

- Heterogeneous and incomplete reporting. **Problem:** several robustness checks are stated without numbers. **Why it matters:** the claim of stability across scales cannot be audited. **Test:** replication with the full set of scales.
- Averaging of window-level statistics. **Problem:** Tables 2 to 5 report averages of window-level R^2 and t -statistics. **Why it matters:** averages of t -statistics have no standard sampling distribution, and window-level R^2 is upward-biased with 180 observations. **Test:** pooled panel regression with stock-by-window fixed effects and clustered errors.
- Multiple testing and specification search over windows, stocks, scales: the share of windows significant at 5 percent is informative, but no correction or sample-splitting is used. **Test:** out-of-sample evaluation on a different month.
- Measurement of β in the second stage. **Problem:** $\hat{\beta}_i$ is estimated with error and can be negative or near zero, so $\log \hat{\beta}_i$ in equation (12) is undefined or very noisy; the treatment is not described. **Test:** errors-in-variables correction or nonlinear least squares on levels.

8 Contradictions and Edge Cases

CRITIQUE The questions below try to break the paper's mechanism, using direct reasoning and calculations on the published numbers.

8.1 Where the mechanism should fail

- Large-tick and small-tick regimes. In the stylised model a tick corresponds to D shares at every level. If the tick is small relative to the price, as for CME at about 322 dollars with a mean spread of 31 cents, queues beyond the best are sparse and a net removal of one queue moves the price by many ticks; then βD exceeds 0.5. The data confirm this: the reviewer's product βD is 3.1 for CME against a cross-sectional median of 0.79, and the R-squared is 35 percent.
- Extreme imbalance. For $|\text{OFI}|$ larger than a few depths, the price crosses levels whose depth is irregular (the paper cites humps and gaps). The linear law would then understate large moves, whereas the paper finds a negative, not positive, quadratic coefficient for trade prices. The tail behaviour of OFI is not examined.
- Zero or negligible depth. As $D \rightarrow 0$ the predicted $\beta = 1/(2D) \rightarrow \infty$, so impact becomes unbounded, although observed price changes are bounded by the spread; the law is valid only where the queue at the best price is a meaningful measure of liquidity.
- Different intervals. At the individual-event scale, the autocorrelation of events (the paper's Figure 1 shows it is small at 10 seconds) is not negligible, and at long horizons mean-reversion of impact would reduce the apparent slope.

8.2 When Proposition 1 gives the wrong exponent

INFERENCE If the event contributions are not independent but have long memory with Hurst-type exponent $b > 1/2$, then $\text{var}[\text{OFI}(T)] \sim N(T)^{2b}$ and $\text{OFI}(T) \sim N(T)^b \sim \text{VOL}(T)^b$, so the artefact exponent becomes b and not $1/2$; with the persistence of order flow documented in the literature (for instance Lillo, Farmer and Mantegna 2003, and the signed-flow memory discussed in Bouchaud et al. 2004) an exponent between 0.6 and 0.8 would be expected at fixed N . If the e_i have heavy tails with index $\alpha < 2$, the scaling is $N^{1/\alpha}$ and the limit is stable, not normal. The paper acknowledges this possibility only through the generalised exponent H in equation (10).

8.3 Simulation of the exponent

INFERENCE The simulation in Figure 3 (generated by the reviewer, with the assumptions listed under the figure) estimates the exponent H from regressions of $\log |\Delta P|$ on $\log \text{VOL}$ in simulated 180-observation intervals in which price changes are exactly βOFI with i.i.d. normal events. When the number of events per interval is constant, the mean estimated exponent is -0.007 and the standard deviation across replications is 0.22: volume carries no information about price moves, although prices are driven only by order flow. As the dispersion of the log event count rises from 0.1 to 0.3 and then 0.5, the mean estimate rises to 0.055, 0.19 and 0.29, and it reaches 0.35 at a dispersion of 1.0. The approximation in equation (10) gives 0.20 at a dispersion of 0.3 and 0.32 at 0.5, which is close for moderate dispersion and overstates for larger dispersion. Two conclusions follow. First, the paper's reported $\hat{H} = 0.18$ is reproduced by a purely mechanical model with moderate variation in activity, so the observation does not discriminate between mechanisms. Second, the result depends on the variation in the number of events across windows, which in real data is dominated by the intraday cycle; a regression done within narrow time-of-day bins should show an exponent closer to zero, a testable prediction that the paper does not make.

8.4 Alternative mechanisms

- Information. News arrival raises both $|\text{OFI}|$ and $|\Delta P|$ without any impact of order flow on prices. The model cannot distinguish this from impact.
- Market-maker repricing. Quote updates can reprice the whole book in response to a signal about the common factor, moving queues and prices simultaneously; a cross-stock factor (such as an index move) would then drive the R-squared. A regression including the contemporaneous imbalance of the index would test this.

9 Further Literature

PAPER The authors place their work among the studies cited in the introduction. **INFERENCE** Table 7 places the paper in a wider set, including later papers that are relevant to the critique. Only works that the reviewer can attribute with confidence are listed.

Table 7. Literature connections

Paper	Shared idea	Relationship	Why it matters
Kyle (1985)	Linear price response to net order flow	Theoretical ancestor: the equilibrium price is linear in total order flow with slope inversely related to liquidity	Gives an information-based reading of β that the paper only alludes to
Hasbrouck (1991)	Informational content of trades	Preceded; trade-based vector autoregression	Benchmark against which the order flow imbalance is compared
Hasbrouck and Seppi (2001)	Price impact depends on contemporaneous liquidity factors	Closest relative among trade-based models	Allows impact to vary through time as in equation (5)
Bouchaud et al. (2004)	Impact of market orders is offset by limit orders	Foundation for the role of limit orders; already in the library	Source of the idea that trades alone mislead
Eisler, Bouchaud and Kockelkoren (2012)	Linear impact of market orders, limit orders, cancellations at the event level	Direct precursor and competitor	Differs by not aggregating events; separates impact by event type
Almgren and Chriss (2000); Obizhaeva and Wang (2013)	Execution cost with linear impact	Application of the paper's input	Motivation: a linear β plugs into these models
Gatheral (2010); Huberman and Stanzl (2004)	No-arbitrage restrictions on impact	Constrain nonlinear permanent impact	Explain why linear impact is favoured
Tóth et al. (2011)	Square-root impact derived from latent liquidity	Contradicts or qualifies the linear law for metaorders	The paper's artefact argument concerns small windows, not metaorders
Cont, Stoikov and Talreja (2010)	Stochastic queueing model of the book	Complementary; same authors, event-level dynamics	A model in which queue depletion moves the price
Gould and Bonart (2016)	Queue imbalance predicts the next tick	Later paper on forecasting	Tests a related variable out of sample
Cont, Cucuringu and Zhang (2023)	Cross-impact of order flow imbalance across stocks	Later extension to many assets	Addresses the single-stock limitation
Kolm, Turiel and Westray (2023)	Deep learning on multi-horizon order flow imbalance	Later application with machine learning	Out-of-sample forecasting with deeper book levels

Source: reviewer's compilation. References are listed in full in Section 14. Relationships are characterised from the contribution of each paper as the reviewer understands it and not from the text of the paper reviewed.

10 Research Gaps

10.1 Paper-level gaps

- **Gap:** contemporaneous association versus prediction. *Why it matters:* execution needs the effect of order flow on future prices. *Existing limitation:* only same-interval regressions. *Way forward:* lagged and out-of-sample tests with multiple horizons.
- **Gap:** decomposition of OFI by event type. *Why it matters:* the claim of equivalence of a market sell and a buy cancellation is an assumption, and impacts per share may differ because cancellations are less informative. *Existing limitation:* the paper aggregates. *Way forward:* Level-2 data with message-level identifiers.
- **Gap:** intercept and asymmetry. *Why it matters:* buys and sells may not have the same impact in falling markets. *Existing limitation:* a symmetric linear law. *Way forward:* separate coefficients for positive and negative imbalance.

10.2 Literature-level gaps

- **Gap:** reconciliation of linear impact at short horizons with square-root impact of metaorders. *Why it matters:* execution models need one coherent object. *Existing limitation:* different data (public quotes versus proprietary parent orders). *Way forward:* models of latent liquidity linking the two.
- **Gap:** causal identification of impact on public data. *Existing limitation:* nearly all public-data studies are associations. *Way forward:* natural experiments.

10.3 Methodological gaps

- **Gap:** second-stage estimation with generated regressors (the half-hourly $\hat{\beta}_i$ enters a regression). *Way forward:* a joint hierarchical model estimated by maximum likelihood or Bayesian methods.
- **Gap:** point process dependence of events. *Way forward:* Hawkes models for signed order flow, as in rough-volatility limits of nearly unstable Hawkes processes (Jaisson and Rosenbaum 2016).

11 Research Opportunities

EXTENSION The five ideas below follow from the weaknesses identified above. Data costs and feasibility statements are the reviewer's estimates and not official quotations.

Idea 1. Removing the mechanical link: queue-only order flow imbalance

RESEARCH QUESTION. How much of the explanatory power of the order flow imbalance survives when events that change the best price are excluded from its construction?

MOTIVATION. Equation (1) includes the price jump in the regressor.

HYPOTHESIS. The R-squared falls substantially, to a level between that of trade imbalance and the original imbalance, but remains positive.

ECONOMIC MECHANISM. Queue changes at constant prices reveal latent supply and demand, which later materialise as price moves.

LITERATURE CONNECTION. Eisler, Bouchaud and Kockelkoren (2012); Cont, Cucuringu and Zhang (2023).

DATA. LOBSTER or Nasdaq TotalView ITCH message data for 100 liquid US stocks over twelve months.

VARIABLES. Mid-price change in ticks; full OFI; queue-only OFI; lagged versions; depth.

METHOD. Window-by-window and pooled panel regressions, with stock and time fixed effects.

IDENTIFICATION STRATEGY. Use lagged queue-only imbalance to predict the next interval's mid-price change; no contemporaneous overlap.

EXPECTED CONTRIBUTION. A clean measure of how much of the claimed impact is accounting.

MAIN RISK. Price-changing events carry most of the signal, leaving little in the queue-only series.

Idea 2. Tick-size pilot as an exogenous shifter of depth

RESEARCH QUESTION. When the minimum tick is raised exogenously, does the product βD stay constant, as the unit-scaling reading predicts, or does it change, as the book-based reading predicts?

MOTIVATION. The $1/D$ law is nearly an identity of units; a change of tick separates the two readings.

HYPOTHESIS. Under unit scaling, βD is invariant to the tick; under a behavioural mechanism, depth rises and βD falls.

ECONOMIC MECHANISM. A larger tick increases queues (time priority has more value) and reduces price levels crossed per unit of flow.

LITERATURE CONNECTION. Cont, Stoikov and Talreja (2010); Hasbrouck and Seppi (2001).

DATA. US tick-size pilot of 2016 to 2018 (treated and control stocks), TAQ and exchange data; the SEC publishes the lists of pilot securities.

VARIABLES. $\hat{\beta}_i$, AD_i , spread, price, volume, group indicators.

METHOD. Difference-in-differences on $\log \hat{\beta}$ and $\log(\hat{\beta} AD)$ with stock and day fixed effects.

IDENTIFICATION STRATEGY. Random assignment of stocks to the treated group.

EXPECTED CONTRIBUTION. A causal test of the depth law.

MAIN RISK. Treated stocks may differ in composition from the 2010 sample; effects may be absorbed by trading-venue changes.

Idea 3. Forecasting content and trading value of order flow imbalance

RESEARCH QUESTION. Does lagged order flow imbalance forecast mid-price changes at horizons of 10 seconds to 5 minutes net of transaction costs?

MOTIVATION. The paper is contemporaneous; practice needs prediction.

HYPOTHESIS. Predictability exists for one or two intervals and disappears net of the half-spread for liquid stocks.

ECONOMIC MECHANISM. Slow-moving liquidity and persistence of order flow imply partial predictability; the spread absorbs most of it.

LITERATURE CONNECTION. Gould and Bonart (2016); Kolm, Turiel and Westray (2023).

DATA. TAQ millisecond data for the S&P 500 over five years, with quotes net of a latency assumption.

VARIABLES. Forward mid-price returns, lagged OFI at several horizons, spread, depth, time of day.

METHOD. Rolling out-of-sample regressions, Campbell and Thompson-type R-squared, and a simple strategy with realistic costs.

IDENTIFICATION STRATEGY. Not causal; prediction evaluated with a rolling or expanding window and purged cross-validation.

EXPECTED CONTRIBUTION. A realistic measure of the value of the signal.

MAIN RISK. Latency and queue-position effects dominate so that the result depends on infrastructure assumptions.

Idea 4. Long memory in order flow and the price-volume exponent

RESEARCH QUESTION. Is the estimated price-volume exponent explained by the long memory of signed order flow rather than by the variation of the event count?

MOTIVATION. Equation (10) and the simulation show that the exponent is a mix of two effects.

HYPOTHESIS. Within narrow time-of-day bins the exponent is close to zero; across bins it is close to the long-memory exponent of the imbalance.

ECONOMIC MECHANISM. Order splitting creates persistence of imbalance, so its variance grows faster than the number of events.

LITERATURE CONNECTION. Lillo, Farmer and Mantegna (2003); Jaisson and Rosenbaum (2016).

DATA. Message data for 100 US stocks; intraday volume profiles.

VARIABLES. OFI over windows of varying length, volume, event count, Hurst-type exponent of the cumulative imbalance.

METHOD. Detrended fluctuation analysis and regression of $\log |\text{OFI}|$ on $\log N$ and $\log \text{VOL}$; Hawkes calibration.

IDENTIFICATION STRATEGY. Within-bin versus between-bin variation, and a simulated Hawkes benchmark.

EXPECTED CONTRIBUTION. Separates the two components of the apparent square-root law.

MAIN RISK. Limited variation of the event count within bins.

Idea 5. Common factor and cross-impact in the regression

RESEARCH QUESTION. How much of the single-stock R-squared is explained by the index-level imbalance, and does adding it change $\hat{\beta}$?

MOTIVATION. A common news or repricing factor can generate the contemporaneous association.

HYPOTHESIS. Controlling for the imbalance of an index ETF lowers the R-squared by a sizeable share for stocks with high index weight.

ECONOMIC MECHANISM. Market makers reprice all constituents together after a common signal.

LITERATURE CONNECTION. Cont, Cucuringu and Zhang (2023); Hasbrouck and Seppi (2001).

DATA. Constituent and ETF (SPY) quotes from TAQ for April 2010 for the same 50 stocks (replication) and a recent year.

VARIABLES. Stock ΔP , stock OFI, SPY OFI and ΔP , index weights.

METHOD. Multivariate regressions with panel structure and a variance decomposition.

IDENTIFICATION STRATEGY. Not causal; the decomposition shows how much is common.

EXPECTED CONTRIBUTION. Quantifies the common-factor share of the association.

MAIN RISK. ETF quote updates themselves contain constituent information, creating a circularity.

11.1 Ranking

Table 8. Ranking of research ideas (scores from 1 to 5, assigned by the reviewer)

Idea	Novelty	Feasibility	Data	Method	Importance	Contribution	Total
1. Queue-only imbalance	4	5	4	4	4	4	25
2. Tick-size pilot	4	4	4	5	4	5	26
3. Forecasting and trading value	3	4	3	4	5	4	23
4. Long memory and exponent	4	3	4	4	3	4	22
5. Common factor and cross-impact	3	4	4	3	3	3	20

Source: reviewer's subjective scoring; no official or empirical basis.

EXTENSION The strongest three are ideas 2, 1 and 3. Idea 2 gives the only design with exogenous variation and answers the key question of whether the depth law is a unit identity; idea 1 is cheap, directly addresses the main weakness, and has an immediate answer; idea 3 has the highest practical value and can be combined with the second stage of idea 1 in a single project.

12 Learning Framework

12.1 Prerequisite concepts

- Structure of a limit order book: queues, price-time priority, tick size, spread, depth.
- Ordinary least squares, heteroskedasticity-consistent (White) and Newey-West standard errors, the meaning of R-squared.
- Law of large numbers and central limit theorem, including random sample size (Anscombe's lemma) and Slutsky's theorem.
- Kyle's linear price impact and the concepts of temporary and permanent impact.

12.2 Core concepts learned

- Order flow imbalance as a single measure of net supply and demand at the best quotes, which treats all order types symmetrically.
- Impact inversely proportional to depth: $\beta \approx c/AD$, with c near 0.45.
- Volume as a derived variable: noisy square-root relation from the central limit theorem.

12.3 Important equations

Equations (1), (4), (5), (7) and (8), and the reviewer's approximation (10), are worth remembering.

12.4 Methods learned

Window-by-window estimation for non-stationary intraday data; two-stage estimation of a parameter that varies in time; National Best Bid and Offer reconstruction from consolidated quotes; the quote test for trade classification; log-log regression for exponents.

12.5 Questions to test understanding

COMPREHENSION

1. How does equation (1) treat a market sell of 500 units and the cancellation of a buy order of 500 units, and why?
2. What does $\beta = 1/(2D)$ mean in the stylised model, and what is ε ?
3. What are the three assumptions of Proposition 1, and what is the role of the law of large numbers in its proof?

ANALYSIS

1. Why is a comparison of the R-squared of order flow imbalance and trade imbalance not a fair horse race?
2. Derive the attenuation formula (10) and explain why $H \rightarrow 0$ when the event count is constant.
3. Why might $\beta \propto 1/D$ hold even if the stylised model is wrong?

RESEARCH

1. Design a test that separates impact from information using an exogenous change in tick size.
2. How would long memory in order flow change Proposition 1?
3. How would you extend the model to the multi-level book?

12.6 Papers to read next

Eisler, Bouchaud and Kockelkoren (2012); Cont, Stoikov and Talreja (2010); Tóth et al. (2011); Cont, Cucuringu and Zhang (2023); Gould and Bonart (2016).

13 Conclusion

13.1 One-sentence thesis

Over short intervals, mid-price changes of liquid equities are, to a good approximation, a linear function of the net change of best-quote queues, with a slope inversely proportional to depth.

13.2 Three most important insights

- Aggregating every order type into the net change in resting liquidity yields a more powerful description of short-run price moves than trade-based variables.
- Impact is a property of the book, not only of the order: the same flow moves a thin book far more than a thick one.
- A square-root relation with volume can emerge mechanically when volume proxies for the event count, so concavity in volume alone is weak evidence on the shape of impact.

13.3 Strongest evidence

The cross-section of 50 stocks in which the order flow imbalance explains 65 percent of the variation of mid-price changes against 32 percent for trade imbalance (Table 4), together with the near unit depth exponent.

13.4 Biggest weakness

The regressor contains the price jumps it is meant to explain, and the design has no exogenous variation, so the numbers measure a contemporaneous association and not impact.

13.5 Most important research gap

A causal and out-of-sample measurement of how order flow imbalance moves future prices.

13.6 Best research opportunity

The exogenous variation provided by the tick-size pilot, combined with a queue-only version of the imbalance.

13.7 Three papers to read next

Eisler, Bouchaud and Kockelkoren (2012); Tóth et al. (2011); Cont, Cucuringu and Zhang (2023).

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